Not Running ●

For the best experience, choose the region closest to you. Then, start the lab to begin.

MARK UNFINISHED

Select the region closest to you

US-East

Keyboard layout: US English (US)

Start lab

SOLUTIONS

Below, you can find solutions for each task. Remember though, that you can follow your own strategy, which may be different from the one explained in the following lab.

Task 1. Is the system infected by Diamorhine?

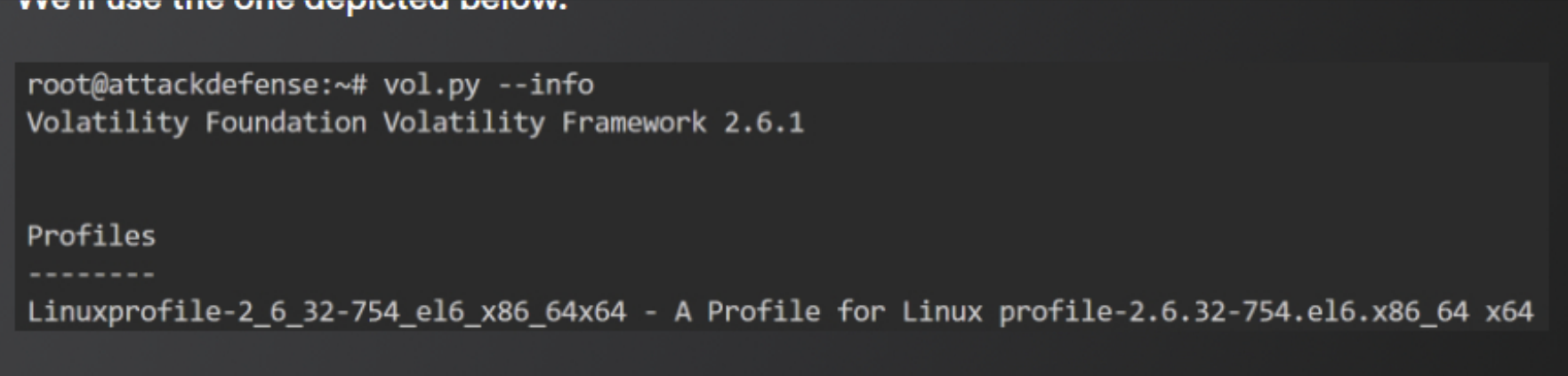
The *linux_check_modules* plugin looks for Loadable Kernel Modules (LKM) that are not listed under */proc/module* but still appear under */sysfs/module* and will output such discrepancies.

The *linux_check_syscall* plugin checks for modifications of the *sys_call_table*. All syscall handler function pointers are listed in the *sys_call_table* array. This plugin compares them with the address specified in the Kernel Symbols Table. In case of a mismatch, the message "hooked" is displayed.

First let's execute the below (inside */root/memory_dump)* to identify all available profiles.

```
vol.py --info
```

We'll use the one depicted below.

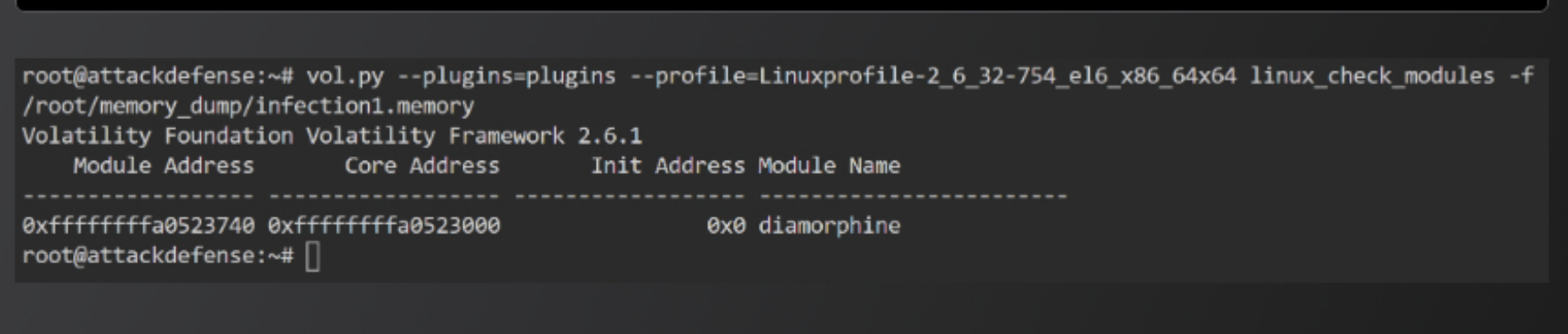


```
root@attackdefense:~# vol.py --info
Volatility Foundation Volatility Framework 2.6.1


Profiles
--------
Linuxprofile-2_6_32-754_el6_x86_64x64 - A Profile for Linux profile-2.6.32-754.el6.x86_64 x64
```

Now, let's start by executing the *linux_check_modules* plugin against *infection1.memory*.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_check_modules
```



```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_check_modules -f
/root/memory_dump/infection1.memory
Volatility Foundation Volatility Framework 2.6.1
    Module Address       Core Address       Init Address Module Name
------------------ ------------------ ------------------ ------------------------
0xffffffffa0523740 0xffffffffa0523000                0x0 diamorphine
root@attackdefense:~# 
```

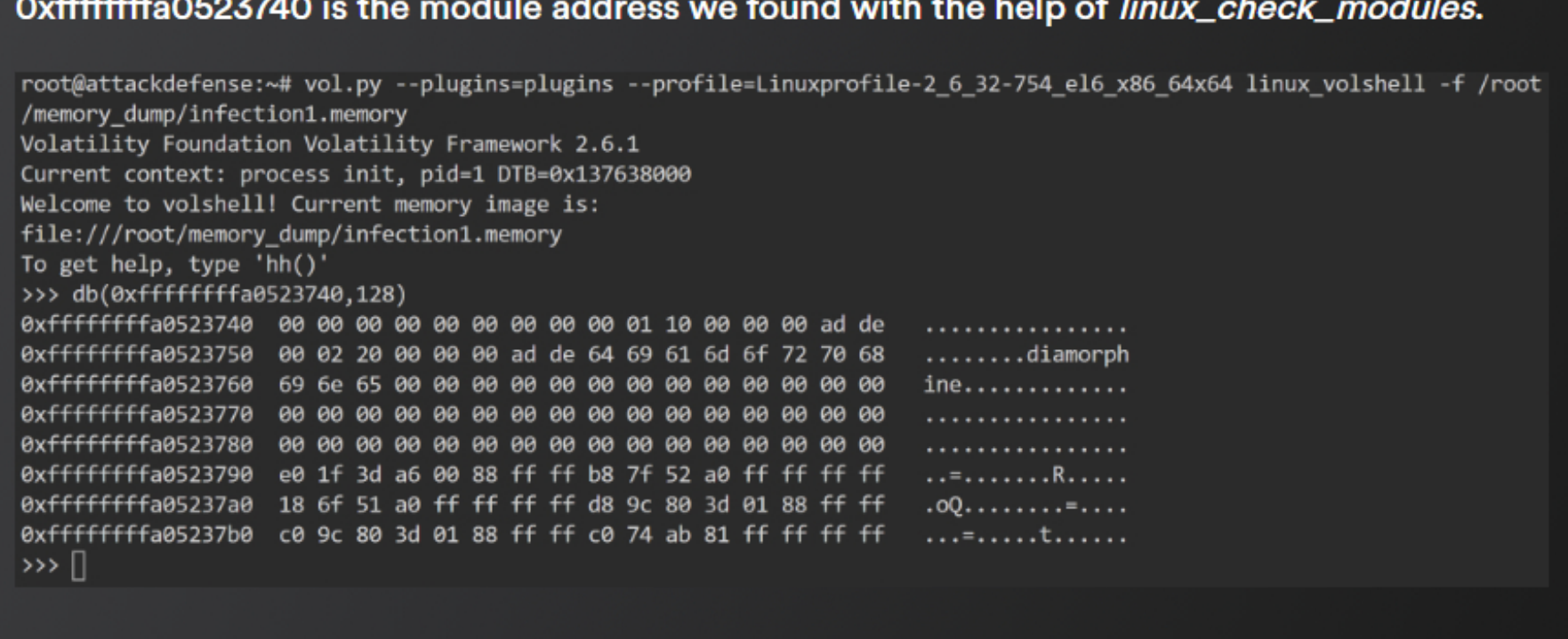
Diamorphine's module name was identified (it was still visible under */sysfs/module*). Let's use *VolShell* to look at the first bytes of that module.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /r
```

Then, we use the
db
command as follows.

```
>>> db(0xffffffffa0523740,128)
```

0xffffffffa0523740 is the module address we found with the help of *linux_check_modules.*

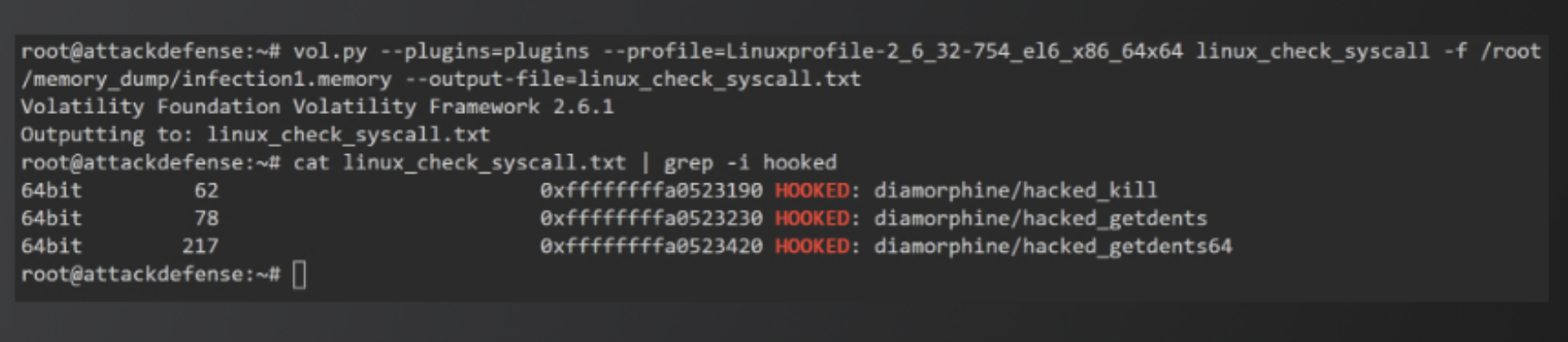


```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /root
/memory_dump/infection1.memory
Volatility Foundation Volatility Framework 2.6.1
Current context: process init, pid=1 DTB=0x137638000
Welcome to volshell! Current memory image is:
file:///root/memory_dump/infection1.memory
To get help, type 'hh()'
>>> db(0xffffffffa0523740,128)
0xffffffffa0523740  00 00 00 00 00 00 00 00 00 01 10 00 00 00 ad de   ................
0xffffffffa0523750  00 02 20 00 00 00 ad de 64 69 61 6d 6f 72 70 68   ........diamorph
0xffffffffa0523760  69 6e 65 00 00 00 00 00 00 00 00 00 00 00 00 00   ine.............
0xffffffffa0523770  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0xffffffffa0523780  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0xffffffffa0523790  e0 1f 3d a6 00 88 ff ff b8 7f 52 a0 ff ff ff ff   ..=.......R.....
0xffffffffa05237a0  18 6f 51 a0 ff ff ff ff d8 9c 80 3d 01 88 ff ff   .oQ........=....
0xffffffffa05237b0  c0 9c 80 3d 01 88 ff ff c0 74 ab 81 ff ff ff ff   ...=.....t......
>>> 
```

Let's now use *linux_check_syscall*, as follows.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_check_syscall
```

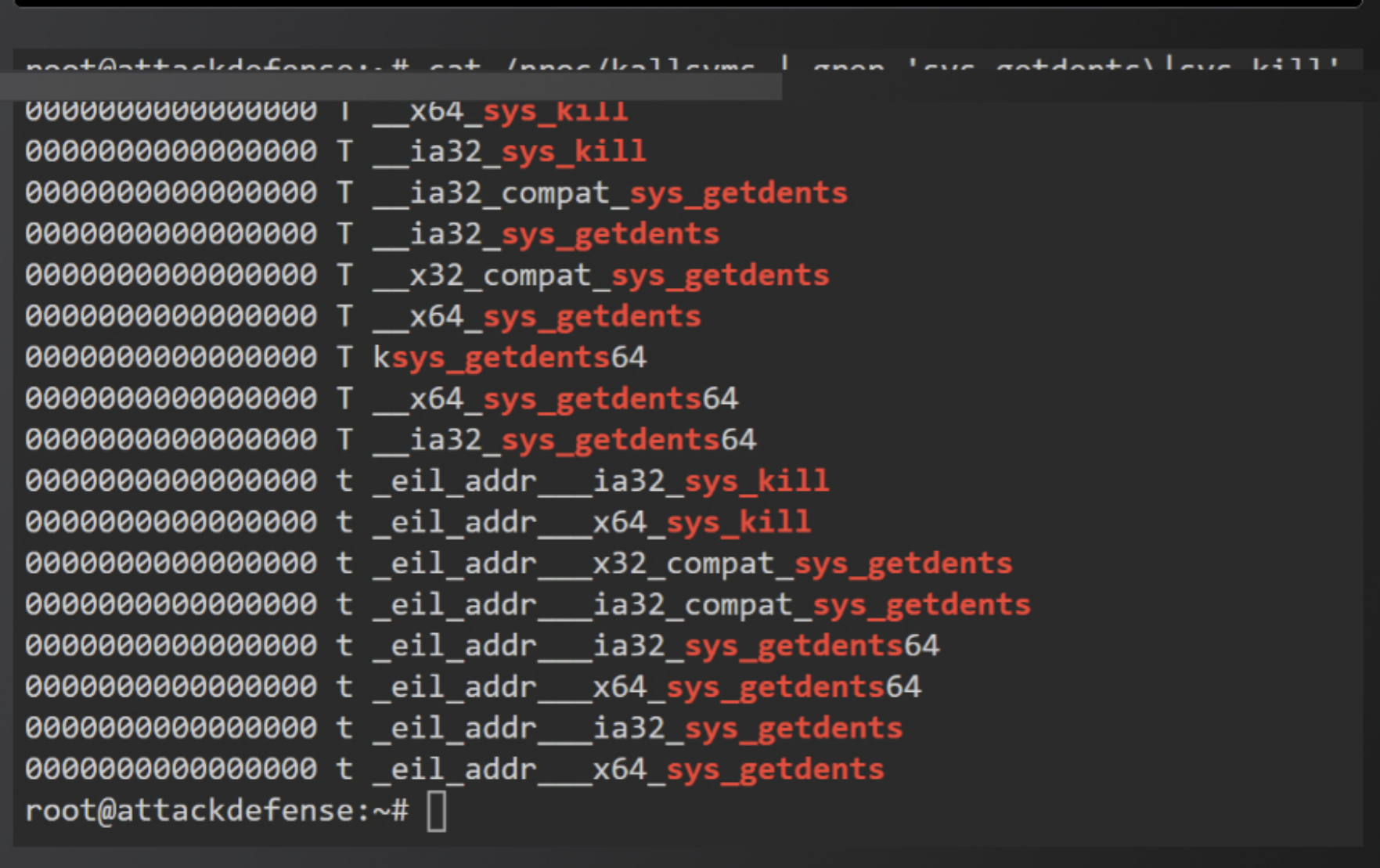
```
cat linux_check_syscall.txt | grep -i hooked
```



```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_check_syscall -f /root
/memory_dump/infection1.memory --output-file=linux_check_syscall.txt
Volatility Foundation Volatility Framework 2.6.1
Outputting to: linux_check_syscall.txt
root@attackdefense:~# cat linux_check_syscall.txt | grep -i hooked
64bit         62                          0xffffffffa0523190 HOOKED: diamorphine/hacked_kill
64bit         78                          0xffffffffa0523230 HOOKED: diamorphine/hacked_getdents
64bit        217                          0xffffffffa0523420 HOOKED: diamorphine/hacked_getdents64
root@attackdefense:~# 
```

linux_check_syscall, identified three hooked syscalls (62, 78 and 217). The machine we are currently at has been disinfected, so we can check the syscall table for the abovementioned syscalls (*sys_kill*, *sys_getdents* and *sys_getdents64*), as follows.

```
cat /proc/kallsyms | grep 'sys_getdents\|sys_kill'
```



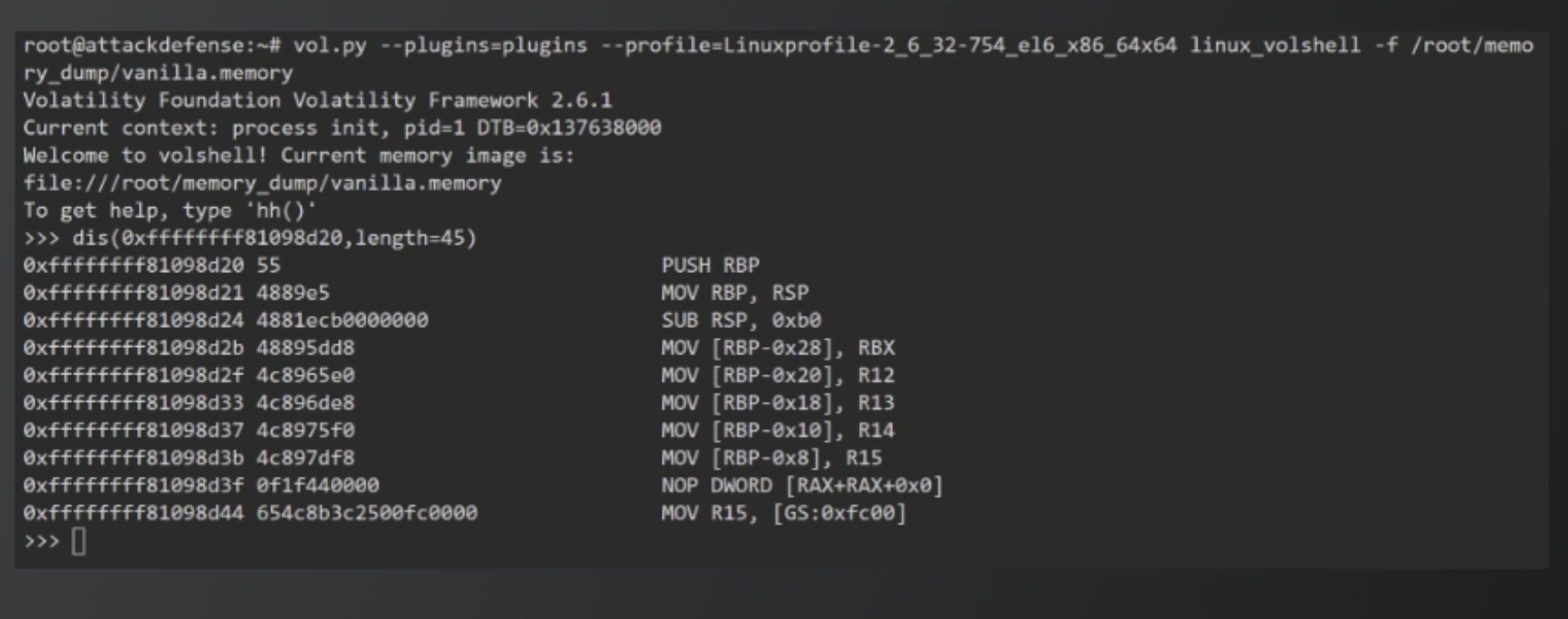
```
root@attackdefense:~# cat /proc/kallsyms | grep 'sys_getdents\|sys_kill'
0000000000000000 T __x64_sys_kill
0000000000000000 T __ia32_sys_kill
0000000000000000 T __ia32_compat_sys_getdents
0000000000000000 T __ia32_sys_getdents
0000000000000000 T __x32_compat_sys_getdents
0000000000000000 T __x64_sys_getdents
0000000000000000 T ksys_getdents64
0000000000000000 T __x64_sys_getdents64
0000000000000000 T __ia32_sys_getdents64
0000000000000000 t _eil_addr___ia32_sys_kill
0000000000000000 t _eil_addr___x64_sys_kill
0000000000000000 t _eil_addr___x32_compat_sys_getdents
0000000000000000 t _eil_addr___ia32_compat_sys_getdents
0000000000000000 t _eil_addr___ia32_sys_getdents64
0000000000000000 t _eil_addr___x64_sys_getdents64
0000000000000000 t _eil_addr___ia32_sys_getdents
0000000000000000 t _eil_addr___x64_sys_getdents
root@attackdefense:~# 
```

Let's look at how *sys_kill* looks like in the pristine system, as follows.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /r
```

Then, we use the *dis* command as follows.

```
dis(0xffffffff81098d20,length=45)
```



```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /root/memo
ry_dump/vanilla.memory
Volatility Foundation Volatility Framework 2.6.1
Current context: process init, pid=1 DTB=0x137638000
Welcome to volshell! Current memory image is:
file:///root/memory_dump/vanilla.memory
To get help, type 'hh()'
>>> dis(0xffffffff81098d20,length=45)
0xffffffff81098d20 55                           PUSH RBP
0xffffffff81098d21 4889e5                       MOV RBP, RSP
0xffffffff81098d24 4881ecb0000000               SUB RSP, 0xb0
0xffffffff81098d2b 48895dd8                     MOV [RBP-0x28], RBX
0xffffffff81098d2f 4c8965e0                     MOV [RBP-0x20], R12
0xffffffff81098d33 4c896de8                     MOV [RBP-0x18], R13
0xffffffff81098d37 4c8975f0                     MOV [RBP-0x10], R14
0xffffffff81098d3b 4c897df8                     MOV [RBP-0x8], R15
0xffffffff81098d3f 0f1f440000                   NOP DWORD [RAX+RAX+0x0]
0xffffffff81098d44 654c8b3c2500fc0000           MOV R15, [GS:0xfc00]
>>> 
```

This is how *sys_kill* looks like in the pristine system.

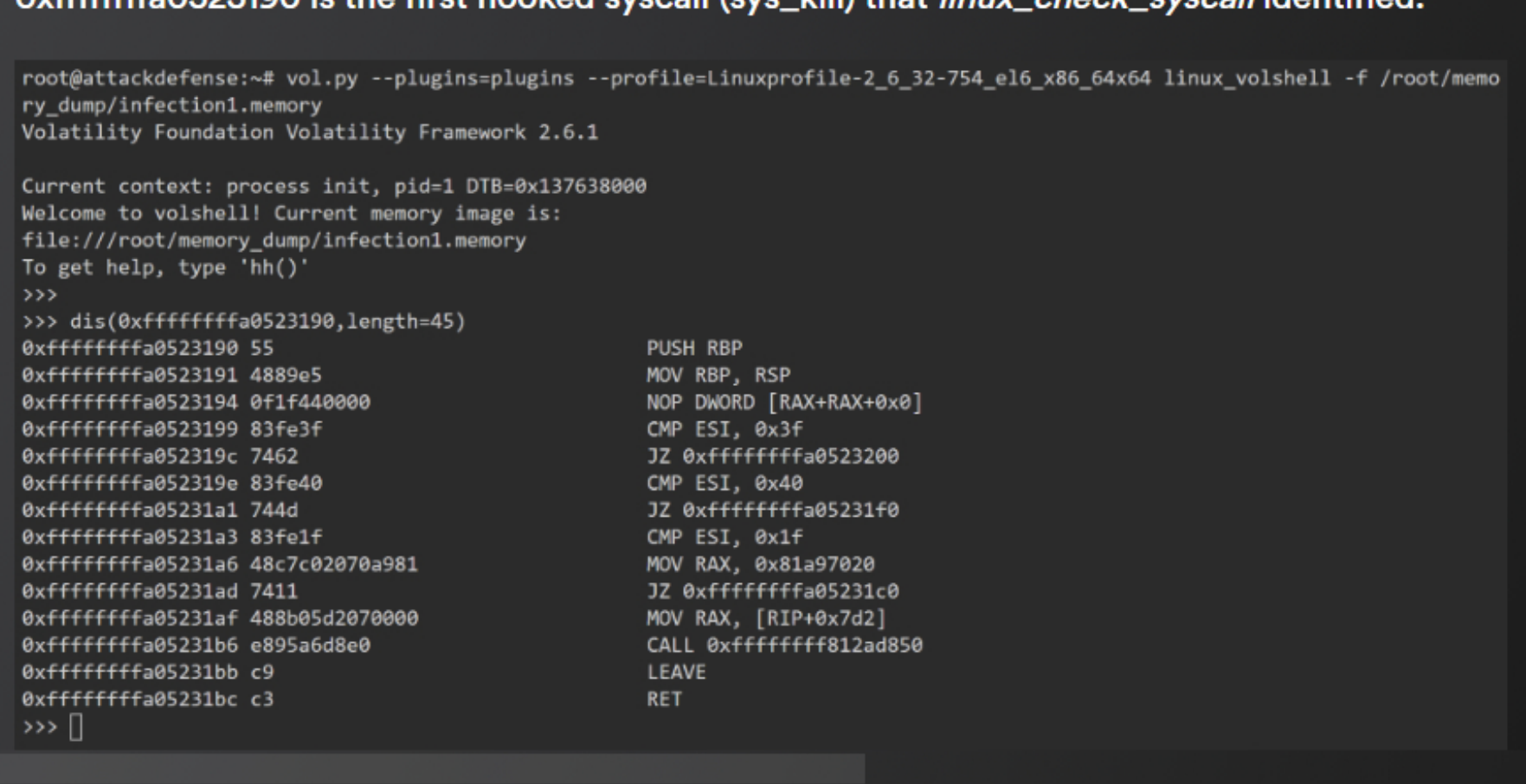
Let's now look at how *sys_kill* looks like in the infected system, as follows.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /r
```

Then, we use the
dis
command as follows.

```
dis(0xffffffffa0523190,length=45)
```

0xffffffffa0523190 is the first hooked syscall (sys_kill) that *linux_check_syscall* identified.



```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /root/memo
ry_dump/infection1.memory
Volatility Foundation Volatility Framework 2.6.1

Current context: process init, pid=1 DTB=0x137638000
Welcome to volshell! Current memory image is:
file:///root/memory_dump/infection1.memory
To get help, type 'hh()'
>>>
>>> dis(0xffffffffa0523190,length=45)
0xffffffffa0523190 55                           PUSH RBP
0xffffffffa0523191 4889e5                       MOV RBP, RSP
0xffffffffa0523194 0f1f440000                   NOP DWORD [RAX+RAX+0x0]
0xffffffffa0523199 83fe3f                       CMP ESI, 0x3f
0xffffffffa052319c 7462                         JZ 0xffffffffa0523200
0xffffffffa052319e 83fe40                       CMP ESI, 0x40
0xffffffffa05231a1 744d                         JZ 0xffffffffa05231f0
0xffffffffa05231a3 83fe1f                       CMP ESI, 0x1f
0xffffffffa05231a6 48c7c02070a981               MOV RAX, 0x81a97020
0xffffffffa05231ad 7411                         JZ 0xffffffffa05231c0
0xffffffffa05231af 488b05d2070000               MOV RAX, [RIP+0x7d2]
0xffffffffa05231b6 e895a6d8e0                   CALL 0xffffffff812ad850
0xffffffffa05231bb c9                           LEAVE
0xffffffffa05231bc c3                           RET
>>> 
```

This is how the hooked *sys_kill* syscall looks like (in the infected system).

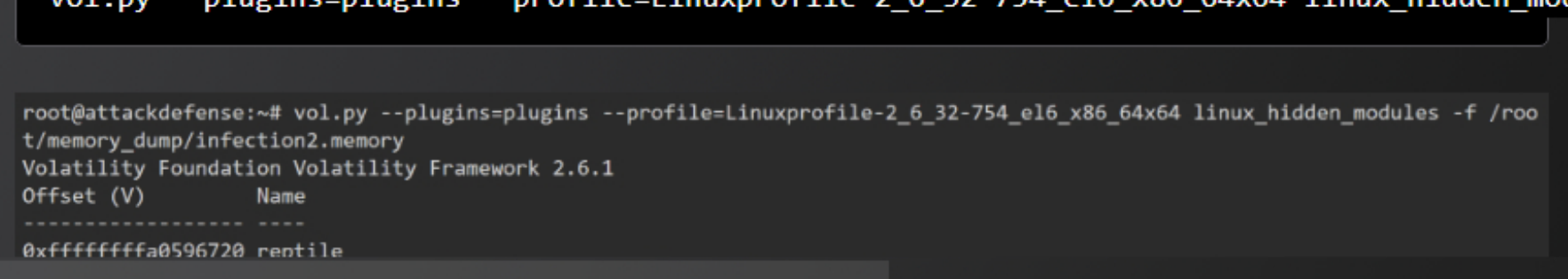
By comparing how the two sycalls (*sys_kill* in the pristine system and *sys_kill* in the infected system) you can see how a hook looks like!

Task 2. Is the system infected by Reptile?

This time we will use *the linux_hidden_modules* plugin. The *linux_check_syscal* plugin can't detect the hooking technique being employed by the Reptile rootkit (since the syscall handler addresses have not been modified).

Among other things, Reptile hooks *fillonedir(), filldir(), filldir64(), compat_fillonedir(), compat_filldir(), compat_filldir64(), __d_lookup().* To hide processes, it hooks *tgid_iter()* and *next_tgid().* To hide network connections, it hooks *tcp4_seq_show* and *udp4_seq_show.*

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_hidden_modules
```

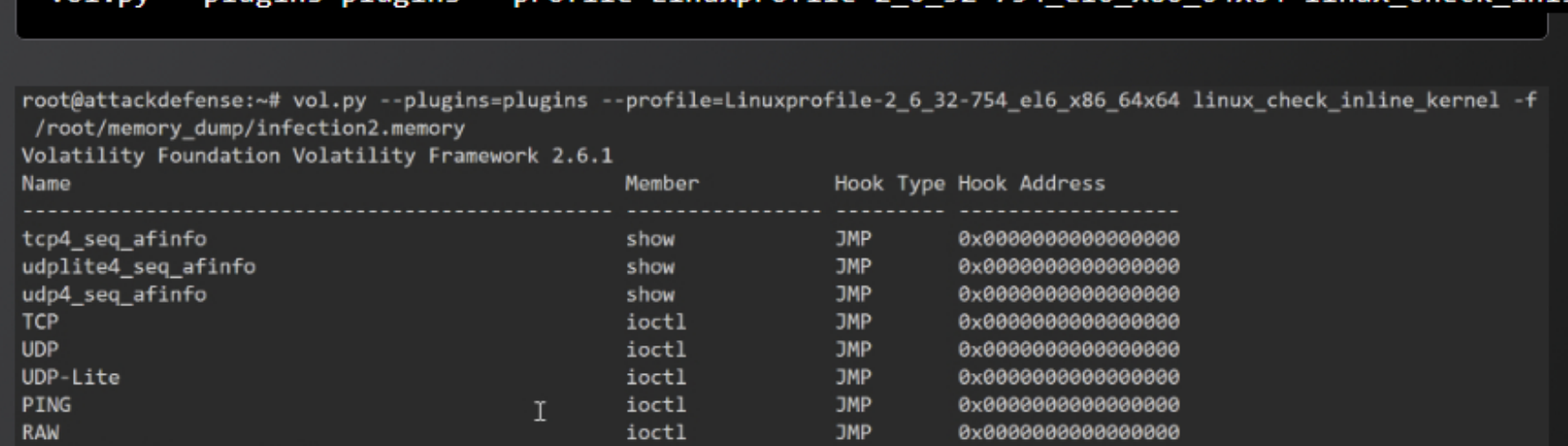


```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_hidden_modules -f /roo
t/memory_dump/infection2.memory
Volatility Foundation Volatility Framework 2.6.1
Offset (V)         Name
------------------ ----
0xffffffffa0596720 reptile
```

The Reptile rootkit hidden module was uncovered by the *linux_hidden_modules* plugin.

Now, let's also check the *linux_check_inline_kernel* plugin. This plugin detects inline hooking. Among other things it checks if the prologue of specific functions in the kernel contains assembly instructions like JMP, CALL or RET and warns the analyst of any functions being hooked.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_check_inline_k
```



```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_check_inline_kernel -f
 /root/memory_dump/infection2.memory
Volatility Foundation Volatility Framework 2.6.1
Name                                           Member           Hook Type Hook Address
---------------------------------------------- ---------------- --------- ------------------
tcp4_seq_afinfo                                show             JMP       0x0000000000000000
udplite4_seq_afinfo                            show             JMP       0x0000000000000000
udp4_seq_afinfo                                show             JMP       0x0000000000000000
TCP                                            ioctl            JMP       0x0000000000000000
UDP                                            ioctl            JMP       0x0000000000000000
UDP-Lite                                       ioctl            JMP       0x0000000000000000
PING                                           ioctl            JMP       0x0000000000000000
RAW                                            ioctl            JMP       0x0000000000000000
root@attackdefense:~# 
```

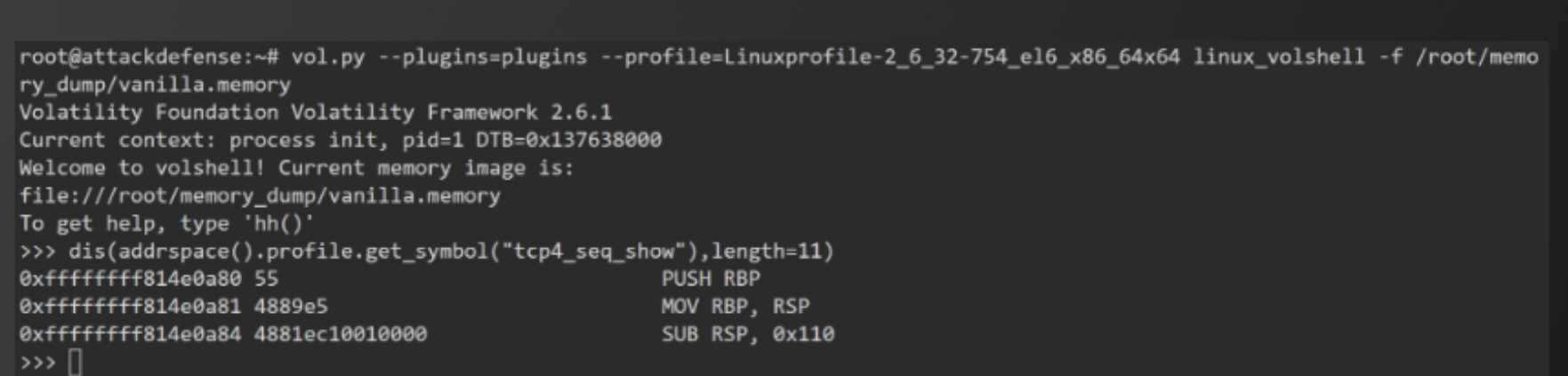
The plugin detected several network-related functions that were patched by the Reptile rootkit.

Let's use VolShell again against both *vanilla.memory* and *infection2.memory* to see how a hooked function(*tcp4_seq_show*) looks like.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /r
```

Then, we use the *dis* command as follows.

```
dis(addrspace().profile.get_symbol("tcp4_seq_show"),length=11)
```

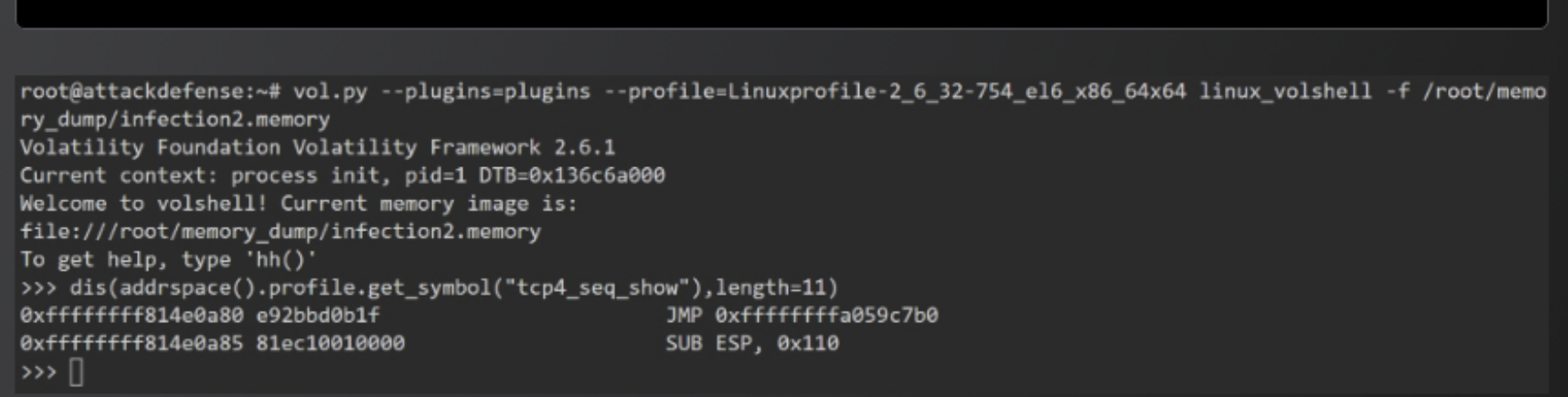


```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /root/memo
ry_dump/vanilla.memory
Volatility Foundation Volatility Framework 2.6.1
Current context: process init, pid=1 DTB=0x137638000
Welcome to volshell! Current memory image is:
file:///root/memory_dump/vanilla.memory
To get help, type 'hh()'
>>> dis(addrspace().profile.get_symbol("tcp4_seq_show"),length=11)
0xffffffff814e0a80 55                           PUSH RBP
0xffffffff814e0a81 4889e5                       MOV RBP, RSP
0xffffffff814e0a84 4881ec10010000               SUB RSP, 0x110
>>> 
```

This is how *tcp4_seq_show* looks like in the pristine system.

Then, we use the *dis* command as follows.

```
dis(addrspace().profile.get_symbol("tcp4_seq_show"),length=11)
```



```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /root/memo
ry_dump/infection2.memory
Volatility Foundation Volatility Framework 2.6.1
Current context: process init, pid=1 DTB=0x136c6a000
Welcome to volshell! Current memory image is:
file:///root/memory_dump/infection2.memory
To get help, type 'hh()'
>>> dis(addrspace().profile.get_symbol("tcp4_seq_show"),length=11)
0xffffffff814e0a80 e92bbd0b1f                   JMP 0xffffffffa059c7b0
0xffffffff814e0a85 81ec10010000                 SUB ESP, 0x110
>>> 
```

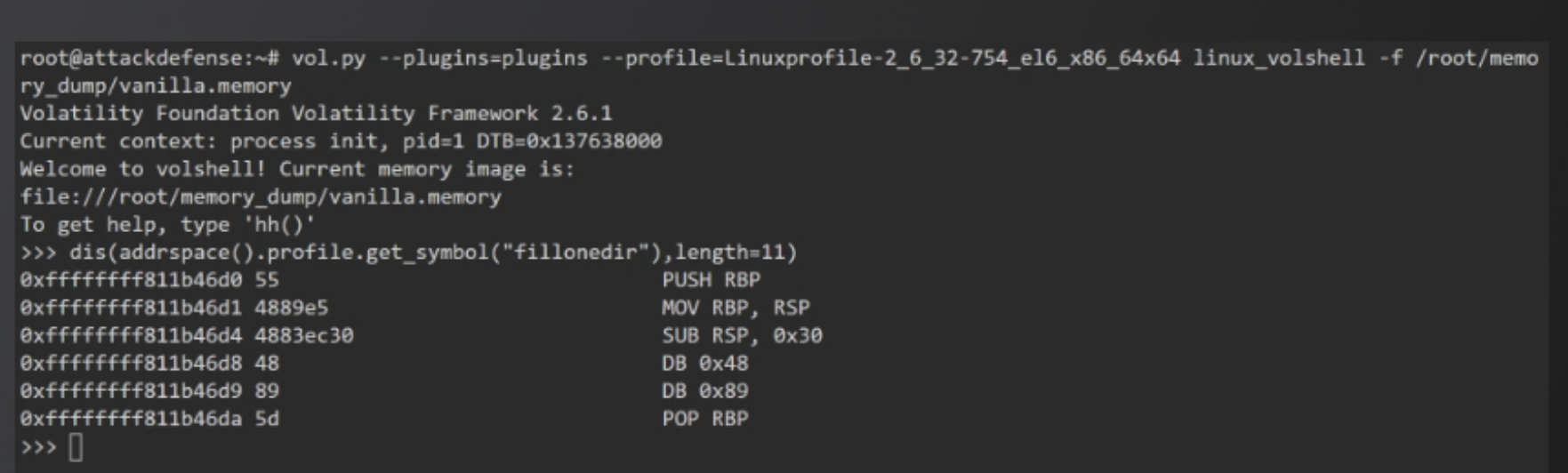
This is how *tcp4_seq_show* looks like in the infected system. It has been patched to jump (JMP) to the Reptile code.

Note: In order to hide directories Reptile also patches the *fillonedir* function. Volatility didn't detect this!

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /r
```

Then, we use the
dis
command as follows.

```
dis(addrspace().profile.get_symbol("fillonedir"),length=11)
```



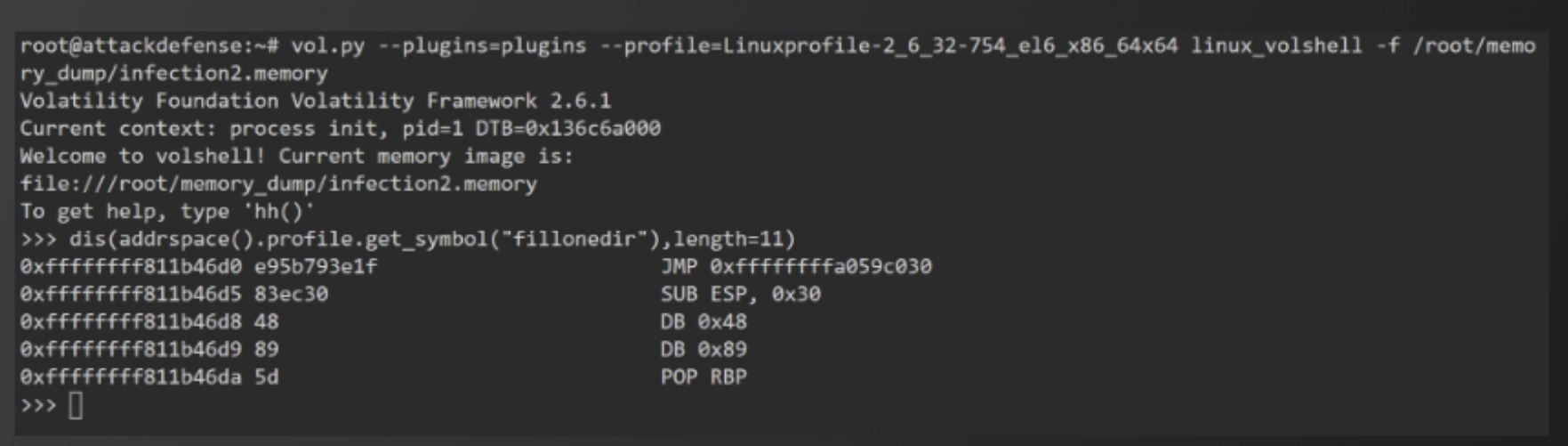
```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /root/memo
ry_dump/vanilla.memory
Volatility Foundation Volatility Framework 2.6.1
Current context: process init, pid=1 DTB=0x137638000
Welcome to volshell! Current memory image is:
file:///root/memory_dump/vanilla.memory
To get help, type 'hh()'
>>> dis(addrspace().profile.get_symbol("fillonedir"),length=11)
0xffffffff811b46d0 55                           PUSH RBP
0xffffffff811b46d1 4889e5                       MOV RBP, RSP
0xffffffff811b46d4 4883ec30                     SUB RSP, 0x30
0xffffffff811b46d8 48                           DB 0x48
0xffffffff811b46d9 89                           DB 0x89
0xffffffff811b46da 5d                           POP RBP
>>> 
```

This is how *fillonedir* looks like in the pristine system.

```
vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /r
```

Then, we use the
dis
command as follows.

```
dis(addrspace().profile.get_symbol("fillonedir"),length=11)
```



```
root@attackdefense:~# vol.py --plugins=plugins --profile=Linuxprofile-2_6_32-754_el6_x86_64x64 linux_volshell -f /root/memo
ry_dump/infection2.memory
Volatility Foundation Volatility Framework 2.6.1
Current context: process init, pid=1 DTB=0x136c6a000
Welcome to volshell! Current memory image is:
file:///root/memory_dump/infection2.memory
To get help, type 'hh()'
>>> dis(addrspace().profile.get_symbol("fillonedir"),length=11)
0xffffffff811b46d0 e95b793e1f                   JMP 0xffffffffa059c030
0xffffffff811b46d5 83ec30                       SUB ESP, 0x30
0xffffffff811b46d8 48                           DB 0x48
0xffffffff811b46d9 89                           DB 0x89
0xffffffff811b46da 5d                           POP RBP
>>> 
```

This is how *fillonedir* looks like in the infected system. It has been patched to jump (JMP) to the Reptile code.

References:

1. http://www.dfir.org/research/android-memory-analysis-DI.pdf
2. https://www.youtube.com/watch?v=oWkOyphlmM8
3. https://github.com/504ensicsLabs/LiME
4. https://www.blackhat.com/presentations/bh-dc-07/Walters/Paper/bh-dc-07-Walters-WP.pdf
5. http://4tphi.net/fatkit/papers/fatkit_journal.pdf
6. http://volatilesystems.blogspot.com/2008/08/pyflagvolatility-team-wins-dfrws.html
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8. https://github.com/halpomeranz/lmg
9. https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf
10. https://github.com/volatilityfoundation/volatility/wiki/Linux-Command-Reference
11. https://github.com/m0nad/Diamorphine
12. https://github.com/f0rb1dd3n/Reptile/
13. https://github.com/h2hconference/2018/
14. https://countuponsecurity.com/2019/10/14/notes-on-linux-memory-analysis-lime-volatility-and-lkms/